Section 7: Hierarchical Human-Inspired Memory Architectures

Literature Review on LLM Context Management

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Overview

This section explores hierarchical memory architectures for LLMs that draw inspiration from human cognitive science, focusing on the multi-tier memory systems (sensory, short-term, long-term) and consolidation mechanisms that have guided recent advances in context management. This is the core theoretical foundation linking cognitive psychology to LLM memory design.

1 7.1 Foundations of Human Memory Theory

1.1 7.1.1 The Atkinson-Shiffrin Multi-Store Model (1968)

The foundational multi-store model of human memory proposed by Atkinson and Shiffrin (1968) established three distinct memory stores:

- **Sensory Memory (SM):** Large capacity, maintains information for hundreds of milliseconds
- **Short-Term Memory (STM):** Severely limited capacity (7±2 items), holds information for seconds to minutes
- **Long-Term Memory (LTM):** Very large capacity, potentially lifetime storage

Key Citations:

- Atkinson, R. C., & Shiffrin, R. M. (1968). Human memory: A proposed system and its control processes. In K. W. Spence & J. T. Spence (Eds.), *The Psychology of Learning and Motivation* (Vol. 2, pp. 89–195). Academic Press.
- Neath, I., & Surprenant, A. M. (2019). 50 years of research sparked by Atkinson and Shiffrin (1968). *Memory & Cognition*.

Important Note: The original framework emphasized *control processes* across all memory systems for both storage and retrieval, not merely the static three-store structure that appears in textbooks. This dynamic processing perspective is highly relevant to modern LLM memory management.

1.2 7.1.2 Baddeley's Working Memory Model (1974, 2000)

Baddeley and Hitch (1974) refined the concept of short-term memory into a multi-component *working memory* system:

- **Central Executive:** Supervisory control system managing attention and cognitive resources
- **Phonological Loop:** Verbal and auditory information processing
  - Phonological store (passive, time-limited holding)
  - Articulatory rehearsal process (active maintenance)
- **Visuospatial Sketchpad:** Visual and spatial information processing
- **Episodic Buffer (added 2000):** Integrated storage of multimodal information linking working memory to LTM

Key Citations:

- Baddeley, A. D., & Hitch, G. (1974). Working memory. In G. H. Bower (Ed.), *The Psychology of Learning and Motivation* (Vol. 8, pp. 47–89). Academic Press.
- Baddeley, A. (2000). The episodic buffer: A new component of working memory? *Trends in Cognitive Sciences, 4*(11), 417–423.

1.3 7.1.3 Miller's Magical Number Seven (1956)

George Miller's seminal paper introduced the concept of "chunking" to explain working memory capacity limitations:

- Immediate memory limited to 7±2 *chunks* of information
- Chunk defined as the largest meaningful unit recognized by the person
- Modern research (Cowan) suggests 4±1 chunks for young adults
- Distinction: Miller's number refers to chunk capacity; Cowan's to active maintenance

Key Citation:

- Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review, 63*(2), 81–97.

1.4 7.1.4 Serial Position Effects: Primacy and Recency

The serial position curve demonstrates differential recall for items based on list position:

- **Primacy Effect:** Enhanced recall for early items (LTM encoding via rehearsal)
- **Recency Effect:** Enhanced recall for recent items (still in STM/working memory)
- Recency cancelled by interfering tasks (15–30 seconds); primacy remains stable
- Originally discovered by Ebbinghaus through self-experimentation

Key Citations:

- Glanzer, M., & Cunitz, A. R. (1966). Two storage mechanisms in free recall. *Journal of Verbal Learning and Verbal Behavior, 5*(4), 351–360.
- Murdock, B. B. (1962). The serial position effect of free recall. *Journal of Experimental Psychology, 64*(5), 482–488.

1.5 7.1.5 Episodic vs. Semantic Memory Distinction

Tulving (1972) distinguished two forms of long-term memory:

- **Semantic Memory:** "Mental thesaurus" for general knowledge, facts, and language
- **Episodic Memory:** Memories of specific events grounded in time and place (temporally dated episodes)
- Tulving (1985) refined this with consciousness levels:
  - Procedural memory: Anoetic (non-knowing)
  - Semantic memory: Noetic (knowing)
  - Episodic memory: Autonoetic (self-knowing)

Key Citations:

- Tulving, E. (1972). Episodic and semantic memory. In E. Tulving & W. Donaldson (Eds.), *Organization of Memory* (pp. 381–403). Academic Press.
- Tulving, E. (1985). Memory and consciousness. *Canadian Psychology/Psychologie Canadienne, 26*(1), 1–12.

Modern Perspective: Recent research suggests episodic and semantic memory exist on a continuum rather than as discrete entities, with shared underlying neural processes.

1.6 7.1.6 The Ebbinghaus Forgetting Curve

Ebbinghaus (1885) established that memory decay follows an exponential function:

- Mathematical form: $R = e^{-t/S}$ where R is retention, t is time, S is memory strength
- Steep initial decline: 50–70% of information lost within 24 hours without reinforcement
- Leveling off at longer intervals
- Debate between exponential vs. power-law decay continues
- Pure forgetting is exponential; heterogeneous material may follow power law

Key Citation:

- Ebbinghaus, H. (1885/1964). *Memory: A contribution to experimental psychology.* Dover.
- Murre, J. M., & Dros, J. (2015). Replication and analysis of Ebbinghaus' forgetting curve. *PLoS ONE, 10*(7), e0120644.

2 7.2 Memory Consolidation and Complementary Learning Systems

2.1 7.2.1 Complementary Learning Systems (CLS) Theory

McClelland, McNaughton, and O'Reilly (1995) proposed the CLS framework explaining why the brain requires two specialized learning systems:

- **Hippocampus:** Sparse, pattern-separated system for rapid episodic memory encoding
- **Neocortex:** Distributed, overlapping system for gradual semantic integration
- **Consolidation Process:** Memories first stored in hippocampus, then gradually transferred to neocortex via repeated reinstatement
- Prevents catastrophic interference while enabling structured knowledge building

Key Citations:

- McClelland, J. L., McNaughton, B. L., & O'Reilly, R. C. (1995). Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *Psychological Review, 102*(3), 419–457.
- Alvarez, P., & Squire, L. R. (1994). Memory consolidation and the medial temporal lobe: A simple network model. *Proceedings of the National Academy of Sciences, 91*(15), 7041–7045.

2.2 7.2.2 Hippocampal Replay and Memory Consolidation

Neural replay mechanisms support memory consolidation through sequential reactivation of experience:

- **Awake Replay:** Frequent reactivation during rest periods and immobility
- **Sleep Replay:** Coordinated hippocampal-cortical replay during slow-wave sleep
- Replay propagates from hippocampus to cortex, strengthening synaptic connections
- Serves multiple functions: consolidation, planning, and memory retrieval

Key Citations:

- Carr, M. F., Jadhav, S. P., & Frank, L. M. (2011). Hippocampal replay in the awake state: A potential substrate for memory consolidation and retrieval. *Nature Neuroscience, 14*(2), 147–153.
- Foster, D. J. (2017). Replay comes of age. *Annual Review of Neuroscience, 40*, 581–602.

2.3 7.2.3 Sleep-Dependent Memory Consolidation

Sleep provides optimal conditions for memory consolidation through coordinated oscillatory activity:

- **Slow Oscillations (SOs):** Coordinate information transfer during slow-wave sleep
- **Spindles and Ripples:** Orchestrate neuronal processing within and across memory circuits

- Active system consolidation redistributes memories to long-term stores
- Alternation of sleep stages facilitates graceful integration of new information with existing knowledge

Key Citations:

- Diekelmann, S., & Born, J. (2010). The memory function of sleep. *Nature Reviews Neuroscience, 11*(2), 114–126.
- Wei, Y., Krishnan, G. P., & Bazhenov, M. (2016). Synaptic mechanisms of memory consolidation during sleep slow oscillations. *Journal of Neuroscience, 36*(15), 4231–4247.
- González-Rueda, A., et al. (2022). A model of autonomous interactions between hippocampus and neocortex driving sleep-dependent memory consolidation. *Proceedings of the National Academy of Sciences, 119*(44), e2123432119.

3 7.3 Cognitive Architectures: Classical Symbolic AI Perspectives

3.1 7.3.1 Production Systems: SOAR and ACT-R

Classical cognitive architectures established foundational memory principles:

- **Production Rules:** If-then statements matching conditions to actions
- **Working Memory:** Relational graph structures (node-edge-value triples)
- **Procedural Memory:** Long-term storage of rules and procedures
- **SOAR Focus:** General AI functionality and efficiency
- **ACT-R Focus:** Detailed modeling of human cognition with specialized modules (visual, memory, motor)

Key Citations:

- Laird, J. E. (2012). *The Soar Cognitive Architecture*. MIT Press.
- Anderson, J. R., et al. (2004). An integrated theory of the mind. *Psychological Review, 111*(4), 1036–1060.

4 7.4 Modern LLM Memory Architectures Inspired by Human Cognition

4.1 7.4.1 CoALA: Cognitive Architectures for Language Agents

Sumers, Yao, Narasimhan, and Griffiths (2023) propose a comprehensive framework organizing language agents along three dimensions:

- **Information Storage:** Working memory (context window) and long-term memory (external stores)

- **Action Space:** Internal actions (reasoning, memory updates) and external actions (tool use, environment interaction)
- **Decision-Making:** Interactive loop with planning and execution phases
- Contextualizes modern language agents within broader AI history
- Identifies actionable directions toward language-based general intelligence

Key Citation:

- Sumers, T. R., Yao, S., Narasimhan, K., & Griffiths, T. L. (2024). Cognitive architectures for language agents. *Transactions on Machine Learning Research.* arXiv:2309.02427.

4.2 7.4.2 Generative Agents: Memory Streams and Reflection

Park et al. (2023) developed computational agents with believable human-like behavior through hierarchical memory:

- **Memory Stream:** Complete natural language record of agent experiences
- **Reflection Mechanism:** Periodic generation of higher-level abstractions from recent memories
  - Triggered when combined importance scores exceed threshold
  - Reflections stored in memory stream alongside observations
- **Retrieval:** Dynamic memory recall based on recency, importance, and relevance
- **Planning:** Future behavior planning informed by retrieved memories
- Enables long-term character development and social interaction

Key Citation:

- Park, J. S., et al. (2023). Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23).* arXiv:2304.03442.

4.3 7.4.3 RAISE: Reasoning and Acting through Scratchpad and Examples

Liu et al. (2024) enhance conversational agents with dual-component memory mirroring human STM/LTM:

- **Dialogue-Level Memory:**
  - Conversation history (maintains context)
  - Scratchpad (intermediate reasoning)
- **Turn-Level Memory:**
  - Examples (query-response pairs for knowledge gaps)
  - Task trajectory (action sequences)

- Fine-tuning method enhances controllability and efficiency
- Particularly effective in domain-specific applications (e.g., real estate sales)

Key Citation:

- Liu, N., et al. (2024). From LLM to conversational agent: A memory enhanced architecture with fine-tuning of large language models. arXiv:2401.02777.

4.4 7.4.4 MemoryBank: Ebbinghaus-Inspired Forgetting

Zhong et al. (2024) implement exponential decay to mimic human forgetting curves:

- **Memory Storage:** Past conversations, summarized events, user portraits
- **Memory Updating:** Exponential decay model based on Ebbinghaus forgetting curve
- **Memory Retrieval:** Vector encoding for similarity-based recall
- **Memory Intensity:** Decaying importance scores over time
- Demonstrated in SiliconFriend chatbot: empathic responses, personality understanding, long-term companionship

Key Citation:

- Zhong, W., Guo, L., Gao, Q., Ye, H., & Wang, Y. (2024). MemoryBank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence, 38*(17). arXiv:2305.10250.

4.5 7.4.5 Think-in-Memory (TiM): Saving Thoughts, Not Just Facts

Liu, Yang, et al. (2023) propose storing post-thinking results rather than raw history:

- **Problem Addressed:** Repeated recall-reason cycles produce biased, inconsistent thoughts
- **Human Inspiration:** People recall thoughts without re-reasoning from scratch
- **Architecture:**
  - Pre-response: Recall relevant thoughts from memory
  - Post-response: Post-think and update memory with new thoughts
- **Memory Operations:** Insert, forget, merge based on similarity (Locality-Sensitive Hashing)
- **Storage:** Hash table for efficient long-term retrieval
- Eliminates repeated reasoning; maintains evolved thought history

Key Citation:

- Liu, L., Yang, X., et al. (2023). Think-in-Memory: Recalling and post-thinking enable LLMs with long-term memory. arXiv:2311.08719.

4.6 7.4.6 Self-Controlled Memory (SCM): Adaptive Memory Integration

Liang, Wang, et al. (2023) introduce memory controllers to prevent noise while extending context:

- **Components:**
  - LLM backbone (instruction-following base model)
  - Memory stream (long-term storage)
  - Memory controller (determines when/how to use memory)
- **Two Memory Types per Segment:**
  - Activation memory (long-term, historical information)
  - Flash memory (short-term, real-time from preceding segment)
- Plug-and-play: No fine-tuning required
- Handles ultra-long texts by selective memory injection

Key Citation:

- Liang, X., Wang, Z., et al. (2023). Enhancing large language model with self-controlled memory framework. arXiv:2304.13343.

4.7 7.4.7 RecallM: Graph-Based Temporal and Relational Memory

Kynoch, Latapie, et al. (2023) use neuro-symbolic graph databases for belief updating:

- **Key Innovation:** Graph database instead of vector database for relational modeling
- **Advantages:**
  - Advanced relationship modeling between concepts
  - Temporal understanding (tracking when information was learned)
  - One-shot belief updating (can revise prior beliefs)
- **Hybrid-RecallM:** Combines graph + vector database for complementary strengths
- Particularly effective for complex relational reasoning and knowledge revision

Key Citation:

- Kynoch, R., Latapie, H., et al. (2023). RecallM: An adaptable memory mechanism with temporal understanding for large language models. arXiv:2307.02738.

4.8 7.4.8 MemGPT: Virtual Context Management as an Operating System

Packer, Fang, et al. (2023) draw inspiration from OS memory hierarchies:

- **Core Concept:** Virtual memory for LLMs—intelligently manage data movement between fast (context window) and slow (external storage) memory
- **Function Calling:** LLM agents use functions to move data between memory tiers
- **Interrupts:** Manage control flow when context limits are reached
- **Applications:**
  - Document analysis far exceeding context window
  - Multi-session chat with long-term memory
- Creates illusion of infinite memory through virtualization

Key Citation:

- Packer, C., et al. (2023). MemGPT: Towards LLMs as operating systems. arXiv:2310.08560.

4.9 7.4.9 RET-LLM: Triplet-Based Knowledge Extraction and Retrieval

Modarressi et al. (2023-2024) implement read-write memory using semantic triplets:

- **Theoretical Foundation:** Davidsonian semantics (event-based meaning representation)
- **Knowledge Format:** Subject-Predicate-Object triplets
- **Memory Properties:**
  - Scalable (grows with new information)
  - Aggregatable (combines related facts)
  - Updatable (revises existing knowledge)
  - Interpretable (human-readable triplets)
- **Controller:** Manages extraction, lookup, and fact-based generation
- Superior performance on temporal question answering and knowledge-intensive tasks

Key Citation:

- Modarressi, A., et al. (2024). RET-LLM: Towards a general read-write memory for large language models. arXiv:2305.14322. (Note: Evolved into MemLLM.)

5 7.5 Transformer Attention as Working Memory Analogue

5.1 7.5.1 Attention Mechanisms and Cognitive Memory Retrieval

Recent research explores parallels between transformer attention and human memory systems:

- **Cue-Based Retrieval:** Attention implements memory retrieval via content-based cues
- **Working Memory Demands:** Attention weights track relational complexity (computational analogue of WM load)
- **Neural Mapping:**
  - Positional encoding aligns with visual cortex representations
  - Attention heads map onto frontoparietal and default-mode networks
- **Hebbian Learning:** Short-term synaptic plasticity can implement transformer-like attention
- **Challenges:** Quadratic complexity and massive comparisons have no clear brain analogue

Key Citations:

- Ryskin, R., et al. (2025). If attention serves as a cognitive model of human memory retrieval, what is the plausible memory representation? arXiv:2502.11469.
- Nikolaev, A., et al. (2025). Aligning transformer circuit mechanisms to neural representations in relational reasoning. *bioRxiv*.
- Kozachkov, L., et al. (2022). Short-term Hebbian learning can implement transformer-like attention. *PNAS Nexus*.

5.2 7.5.2 TransformerFAM: Feedback Attention as Working Memory

Recent work explicitly models feedback loops in attention to simulate working memory:

- Feedback attention mechanisms enable iterative refinement
- Recurrent processing within attention layers
- Closer alignment with cognitive theories of active maintenance

Key Citation:

- Irie, K., et al. (2024). TransformerFAM: Feedback attention is working memory. arXiv:2404.09173.

6 7.6 Episodic vs. Semantic Memory in LLMs

6.1 7.6.1 The Distinction Applied to LLMs

Human episodic/semantic memory maps onto LLM memory types:

- **Semantic Memory (LLMs):** Parametric knowledge acquired during pre-training/fine-tuning
  - Stored implicitly in model weights

- General truths and common knowledge
  - Mirrors human semantic memory for factual information
- **Episodic Memory (LLMs):** Context-specific, temporally grounded events
  - Specific interactions, conversations, events
  - Linked to time, place, and situational context
  - Personal, experiential memories
- **Blurred Boundaries:** Modern research views these as a continuum, not discrete categories

6.2 7.6.2 Challenges and Solutions

Standard LLMs excel at semantic knowledge but struggle with episodic memory:

- **Challenges:**
  - Difficulty with once-presented information over long timescales
  - Poor performance on sequence order recall and spatio-temporal relationships
  - Multiple related events cause interference
- **Memory-Augmented Solutions:**
  - External episodic memory systems (EM-LLM, etc.)
  - Explicit temporal tagging and event segmentation
  - Structured retrieval combining recency, relevance, and importance

Key Citations:

- Ning, Y., et al. (2025). Towards large language models with human-like episodic memory. *Trends in Cognitive Sciences*.
- Kwon, M., et al. (2024). Assessing episodic memory in LLMs with sequence order recall tasks. arXiv preprint.
- Sun, Y., et al. (2025). Position: Episodic memory is the missing piece for long-term LLM agents. arXiv:2502.06975.

7 7.7 Continual Learning and Long-Term Memory as Parametric Updates

7.1 7.7.1 Catastrophic Forgetting in LLMs

When neural networks learn new tasks, they often forget previously learned information:

- **Definition:** Catastrophic interference where new training disrupts prior knowledge
- **Mechanisms in LLMs:**
  - Gradient interference in attention weights

- Representational drift in intermediate layers
- Loss landscape flattening around prior task minima

- **Severity:** More severe in larger models (1B–7B parameters); 15–23% of lower-layer attention heads show severe disruption
- **Relation to Memory:** Analogous to failure of memory consolidation in LTM

Key Citations:

- Luo, Y., et al. (2023). An empirical study of catastrophic forgetting in large language models during continual fine-tuning. arXiv:2308.08747.
- Chen, Z., et al. (2025). Catastrophic forgetting in LLMs: A comparative analysis across language tasks. arXiv:2504.01241.
- Zhang, T., et al. (2025). Mechanistic analysis of catastrophic forgetting in large language models during continual fine-tuning. arXiv:2601.18699.

7.2 7.7.2 Mitigation Strategies

Approaches to reduce catastrophic forgetting parallel biological consolidation:

- **Parameter-Efficient Fine-Tuning (PEFT):** LoRA, adapters, prefix tuning
  - Freeze base weights, train small adapter modules
  - Limited success: LoRA alone does not prevent forgetting
- **General Instruction Tuning:** Broad task coverage during initial training reduces later forgetting
- **Continual Learning Techniques:**
  - Replay (experience replay buffers)
  - Regularization (Elastic Weight Consolidation)
  - Dynamic architectures (progressive networks)
- **Promising Models:** Phi-3.5-mini shows minimal forgetting while maintaining learning capabilities

7.3 7.7.3 LoRA: Low-Rank Adaptation for Knowledge Updates

LoRA enables efficient adaptation of LLMs without full re-training:

- **Mechanism:** Freeze pre-trained weights, inject trainable low-rank matrices ($\Delta W = BA$)
- **Benefits:**
  - 10,000× fewer trainable parameters than full fine-tuning
  - 3× reduction in GPU memory requirements
  - Rapid task switching (swap LoRA weights at inference)

- **Limitations:**
  - Knowledge primarily from pre-training, not fine-tuning
  - Does not fully mitigate catastrophic forgetting
  - Acts as "delta" to original weights—compact, targeted changes
- **Relation to LTM:** LoRA updates approximate minimal perturbations to parametric memory

Key Citation:

- Hu, E. J., et al. (2022). LoRA: Low-rank adaptation of large language models. In *Proceedings of ICLR 2022*. arXiv:2106.09685.

8 7.8 Sleep-Inspired Consolidation for Artificial Neural Networks

8.1 7.8.1 Biological Inspiration: Offline Processing

Sleep enables memory consolidation without sensory input interference:

- Memory networks repeatedly reactivated during sleep
- Slow oscillations, spindles, and ripples coordinate consolidation
- Hippocampal-neocortical interactions transfer episodic to semantic memory
- Graceful integration of new knowledge with existing schemas

8.2 7.8.2 NeuroDream: Explicit Dream Phases for ANNs

Tutuncuoglu (2024) introduces sleep-like consolidation into neural network training:

- **Dream Phase:** Model disconnects from input data, engages in internal simulations
- **Latent Rehearsal:** Replay stored embeddings and learned dynamics
- **Abstraction:** Extract patterns from earlier experiences without raw data re-exposure
- **Benefits:**
  - Mitigates catastrophic forgetting
  - Improves generalization
  - Enables recontextualization of learned knowledge
- **Contrast with Standard ANNs:** Typical networks rely only on gradient descent during active training

Key Citation:

- Tutuncuoglu, B. T. (2024). NeuroDream: A sleep-inspired memory consolidation framework for artificial neural networks. SSRN Working Paper.

8.3 7.8.3 Computational Models of Hippocampal-Neocortical Consolidation

Simulations demonstrate autonomous offline learning:

- **Hippocampal Replay:** Reactivates recent experiences during simulated sleep
- **Neocortical Integration:** Builds structured representations via interleaved replay
- **Sleep Stage Alternation:** Different stages (REM, NREM) serve complementary consolidation roles
- Provides mechanistic account of how offline processing avoids interference

Key Citation:

- González-Rueda, A., et al. (2022). A model of autonomous interactions between hippocampus and neocortex driving sleep-dependent memory consolidation. *Proceedings of the National Academy of Sciences, 119*(44), e2123432119.

9 7.9 Infinite Context and Unbounded Memory Architectures

9.1 7.9.1 The Challenge: Quadratic Complexity of Attention

Transformer attention exhibits $O(n^2)$ memory and computation:

- For 500B model, batch size 512, context 2048: 3TB memory footprint
- Context of 20,000 tokens requires 126GB for KV cache
- Scaling to millions of users and trillion-parameter models is infeasible

9.2 7.9.2 Infini-Attention: Compressive Memory for Infinite Context

Google (2024) introduces compressive memory with linear attention:

- **Mechanism:** Additional compressive memory layer alongside standard attention
- **Scaling:** Handles 1M+ token contexts with 114× less memory
- **Capabilities:**
  - Retrieve random number inserted in 1M-token text
  - Summarize 500K-token books effectively
- **Performance:** Lower perplexity than baseline transformers on long-context tasks

Key Citation:

- Munkhdalai, T., et al. (2024). Leave no context behind: Efficient infinite context transformers with Infini-attention. arXiv:2404.07143.

9.3 7.9.3 EM-LLM: Human-Inspired Episodic Memory for Infinite Context

EM-LLM (2025) integrates episodic memory and event cognition for unbounded context:

- **Event Segmentation:** Bayesian surprise + graph-theoretic boundary refinement
- **Two-Stage Retrieval:**
  - Similarity-based retrieval
  - Temporally contiguous retrieval
- **No Fine-Tuning:** Plug-and-play integration
- **Performance:**
  - Surpasses full-context models on most tasks
  - Successfully retrieves across 10M tokens
  - Computationally infeasible for standard transformers at this scale

Key Citation:

- Hu, Z., et al. (2025). EM-LLM: Human-inspired episodic memory for infinite context LLMs. arXiv:2407.09450.

9.4 7.9.4 Cognitive Workspace: Extended Mind for LLMs

Inspired by Baddeley, Clark's extended mind thesis, and distributed cognition:

- **Theoretical Foundations:**
  - Baddeley's working memory model
  - Clark & Chalmers' extended mind thesis
  - Hutchins' distributed cognition framework
- **Key Concept:** Transcends traditional RAG by emulating human external memory use
- Manages memory as computational resource beyond static retrieval

Key Citation:

- Wang, Y., et al. (2025). Cognitive workspace: Active memory management for LLMs. arXiv:2508.13171.

10 7.10 Comparative Analysis: Human vs. LLM Memory

10.1 7.10.1 Structural and Functional Differences

Property	Human Memory	LLM Memory
Sensory Memory	Large capacity, 100–500ms retention	API request/prompt input (immediate processing)

Property	Human Memory	LLM Memory
Short-Term/Working Memory	7±2 chunks (Miller) or 4±1 (Cowan); seconds to minutes	Context window (512–128K tokens); no inter-session persistence
Long-Term Memory	Potentially unlimited capacity; lifetime storage; dynamic, reconstructive	Parametric (fixed post-training) + Non-parametric (external databases/RAG)
Memory Formation	Encoding → Consolidation → Storage (hippocampus → neocortex)	Training (parametric) or storage in external memory (non-parametric)
Consolidation	Sleep-dependent; hippocampal replay; gradual neocortical integration	Absent in standard LLMs; experimental in NeuroDream, offline consolidation research
Forgetting	Exponential decay (Ebbinghaus); interference; retrieval failure	No intrinsic forgetting; catastrophic interference during fine-tuning
Episodic Memory	Temporally and spatially grounded events; autonoetic consciousness	External memory systems (EM-LLM, Generative Agents); context-dependent
Semantic Memory	General knowledge; facts; noetic consciousness	Parametric knowledge from pre-training; static until re-training
Retrieval Mechanism	Cue-based; reconstructive; influenced by emotion, context, biases	Attention-based (parametric) or similarity search (non-parametric); deterministic
Statefulness	Continuous, persistent identity and memory across time	Stateless by default; each interaction independent unless memory systems added
Adaptability	Highly adaptive; context-sensitive; personal biases shape recall	Static parametric knowledge; adaptable via external memory or fine-tuning
Computational Cost	Biological efficiency; parallel processing; low energy	Quadratic attention complexity; high energy consumption; memory bottlenecks
Updating Knowledge	Continuous learning; synaptic plasticity; no catastrophic forgetting	Requires re-training or PEFT (LoRA); catastrophic forgetting during continual learning

Table 1: Comparative Properties of Human Memory vs. LLM Memory Systems

10.2 7.10.2 Key Insights from the Comparison

- **Statelessness vs. Continuity:** LLMs are fundamentally stateless; memory-augmented systems approximate human-like continuity
- **Parametric vs. Reconstructive:** LLMs encode patterns in parameters (fixed); humans reconstruct memories dynamically
- **Consolidation Gap:** Sleep-dependent consolidation is central to human memory but absent in standard LLMs
- **Catastrophic Forgetting:** LLMs lack biological mechanisms (complementary learning systems) that prevent interference
- **Episodic Memory Challenge:** LLMs struggle with temporally grounded, context-rich episodic recall
- **Scalability:** Quadratic attention complexity limits LLM context scaling; human memory scales gracefully

11 7.11 Synthesis and Future Directions

11.1 7.11.1 Summary of Hierarchical Memory Design Principles

Human-inspired LLM memory architectures converge on several key principles:

1. **Multi-Tier Memory:** STM (context window) + LTM (external storage) + intermediate memory (e.g., episodic buffer)
2. **Selective Consolidation:** Not all information should persist; importance-weighted storage (Generative Agents, MemoryBank)
3. **Temporal Awareness:** Timestamps, decay functions, temporal ordering critical for episodic memory (RecallM, MemoryBank)
4. **Structured Retrieval:** Combine recency, relevance, and importance (Generative Agents, EM-LLM)
5. **Dynamic Updates:** Memory systems must support belief revision and knowledge updates (RecallM, RET-LLM)
6. **Reflection and Abstraction:** Higher-order processing generates summaries and insights (Generative Agents, Think-in-Memory)
7. **Controlled Integration:** Memory controllers prevent noise accumulation (SCM, MemGPT)
8. **Event Segmentation:** Chunk experiences into meaningful episodes (EM-LLM, CoALA)

11.2 7.11.2 Open Research Questions

- **Biological Plausibility:** Can LLMs implement truly brain-like consolidation mechanisms?
- **Catastrophic Forgetting:** How to achieve continual learning without catastrophic interference?
- **Episodic Richness:** Can external memory systems capture the full richness of human episodic memory?
- **Scalability:** How to achieve human-like memory efficiency at scale (beyond quadratic attention)?
- **Sleep-Inspired Learning:** Will offline consolidation phases become standard in LLM training pipelines?
- **Unified Architectures:** Can a single framework integrate parametric, episodic, and semantic memory seamlessly?

11.3 7.11.3 Implications for Context Management

Hierarchical memory architectures transform context management from a purely technical challenge to a cognitively inspired design space:

- Context windows are working memory, not the entirety of memory
- Effective context management requires consolidation, retrieval, and forgetting mechanisms
- Episodic and semantic memory serve complementary roles in long-horizon tasks
- Biological memory systems provide blueprints for scalable, adaptive LLM memory

Sources and Web Search Citations

This refined outline was developed through extensive web searches conducted on January 27, 2026. All claims are supported by peer-reviewed research and preprints. Key sources include foundational cognitive science papers (Atkinson & Shiffrin 1968, Tulving 1972/1985, Miller 1956, Ebbinghaus 1885), neuroscience research on memory consolidation (McClelland et al. 1995, Carr et al. 2011, González-Rueda et al. 2022), and recent LLM memory architectures (CoALA, Generative Agents, MemoryBank, Think-in-Memory, MemGPT, EM-LLM, RecallM, RET-LLM, SCM, RAISE).

Web Search References:

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- Baddeley's working memory: `https://en.wikipedia.org/wiki/Baddeley's_model_of_working_memory`
- Complementary Learning Systems: `https://pubmed.ncbi.nlm.nih.gov/22141588/`
- Ebbinghaus forgetting curve: `https://en.wikipedia.org/wiki/Forgetting_curve`
- CoALA: `https://arxiv.org/abs/2309.02427`

- Generative Agents: https://arxiv.org/abs/2304.03442
- MemoryBank: https://arxiv.org/abs/2305.10250
- RAISE: https://arxiv.org/abs/2401.02777
- Think-in-Memory: https://arxiv.org/abs/2311.08719
- SCM: https://arxiv.org/abs/2304.13343
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- SOAR and ACT-R: https://en.wikipedia.org/wiki/Soar_(cognitive_architecture)
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- Episodic memory in LLMs: https://www.sciencedirect.com/science/article/abs/pii/S136466132500179